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Course Name

Computer Programming I

Topic: Programming Languages and their Classifications

Week 2

BULLET POINT SUMMARY

This week, some of the learning points include:

- Programming languages translate human instructions into computer-understandable code, enabling communication and creation.
- High-level languages (Python, Java) are user-friendly, while low-level languages (Assembly) offer fine-grained control but require more expertise.
- Inside the machine, machine language, which consists of binary instructions, is the core language computers understand.
- Programming paradigms like procedural (C), object-oriented (Java), functional (Haskell), and scripting (JavaScript) offer various ways to solve problems.
- Specialised languages like SQL for databases optimise development for specific tasks.
- Compiled languages (C++) are translated to machine code beforehand for faster execution, while interpreted languages (Python) are executed line-by-line.
- Web Development Trio: HTML structures content, CSS styles it, and JavaScript adds interactivity for building websites.
- Languages adapt and new ones emerge, offering diverse tools for developers to tackle challenges and innovate.



PERSONAL REFLECTION

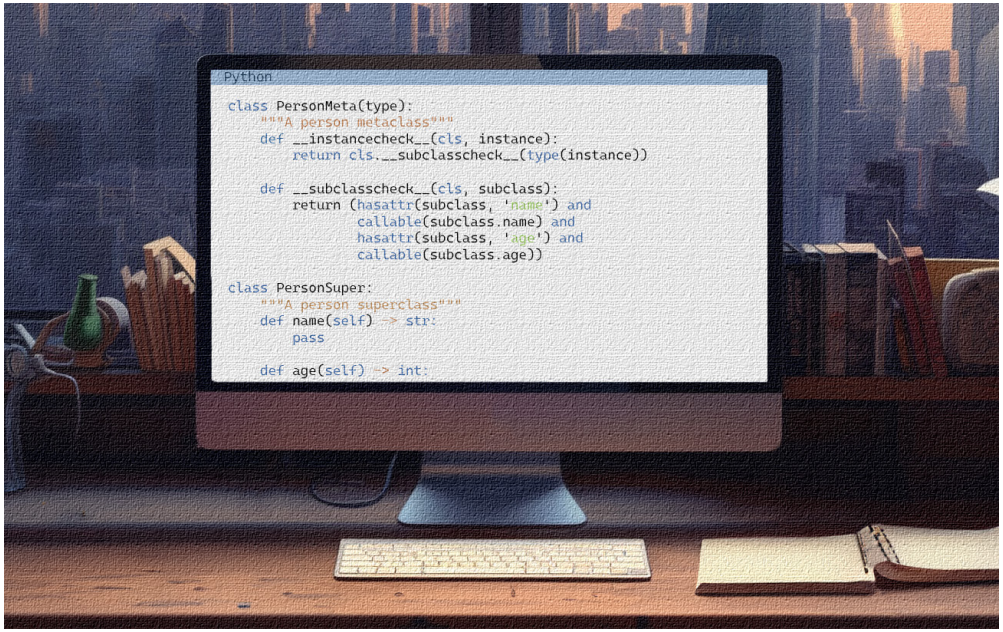
As technology advances, new programming languages and paradigms emerge.

- How do you stay informed about these advancements?
- How can you cultivate a learning mindset to adapt and embrace the evolving landscape of programming languages and their applications?

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Key references: Principles and Practice of Declarative Programming. (1999). Germany: Springer.

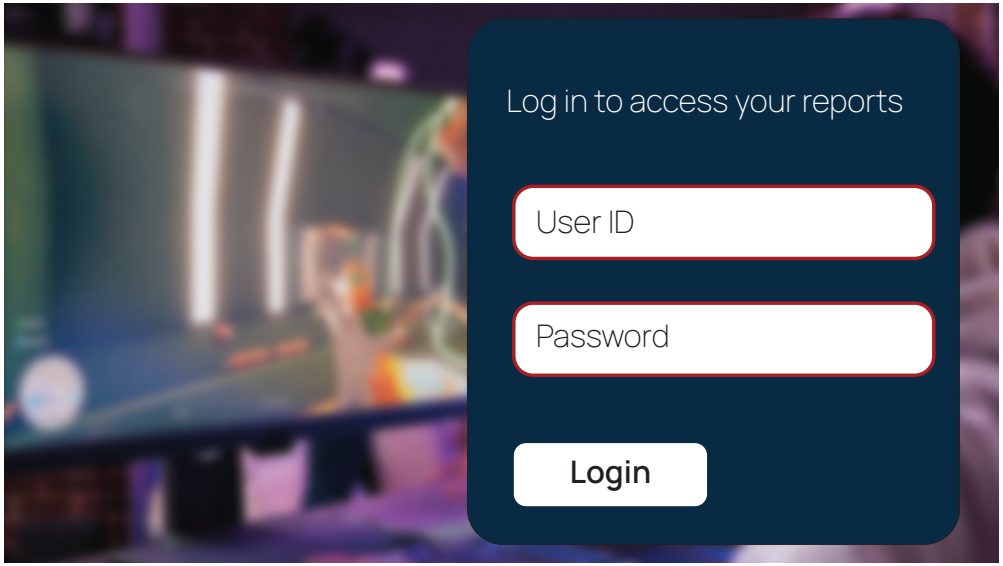
CALL TO ACTION



Imagine you are interested in building a mobile app showcasing your skills but have limited time and resources.

- How would you choose the right language and development tools based on your skillset, project scope, and desired learning outcomes?
- What strategies would you employ for efficient development, testing, and deployment within a budget and timeframe?

SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON



- Critical Thinking
- Problem-Solving:
- Versatility
- Adaptability:
- Language Selection

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Key references:

Braßel, B. (2012). Implementing Functional Logic Programs: By Translation Into Purely Functional Programs. Germany: KS Omniscriptum Publishing.